

4. ReactJS-HOL

Objectives

- Explain the need and Benefits of component life cycle
- Identify various life cycle hook methods
- List the sequence of steps in rendering a component

In this hands-on lab, you will learn how to:

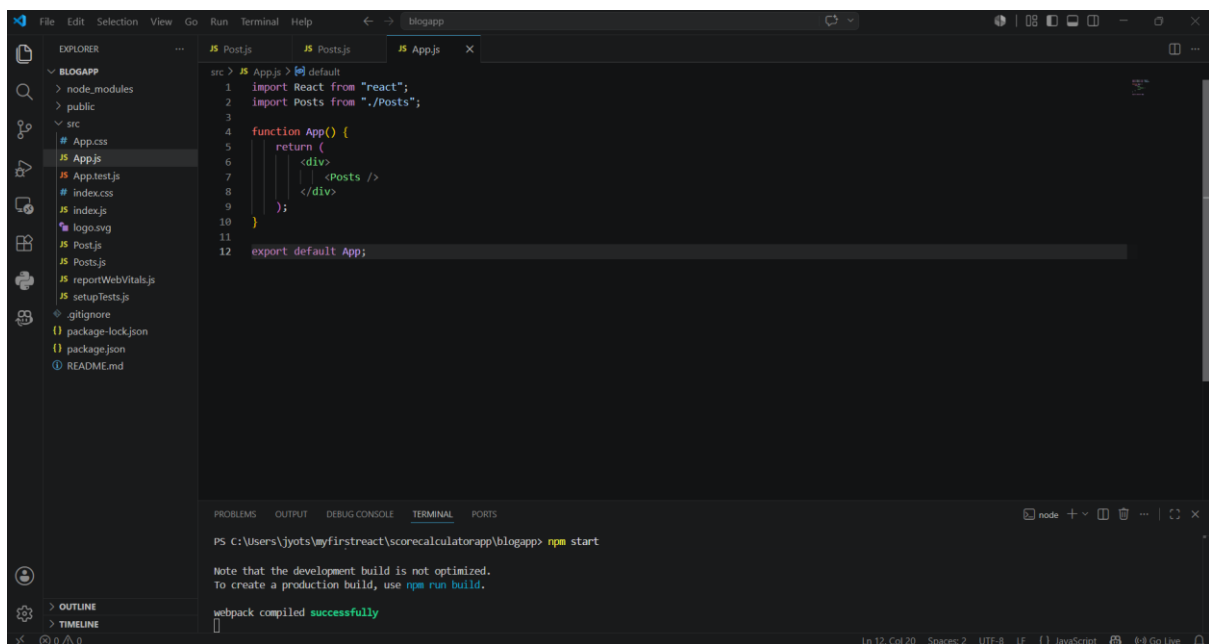
- Implement componentDidMount() hook
- Implementing componentDidCatch() life cycle hook.

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

App.js

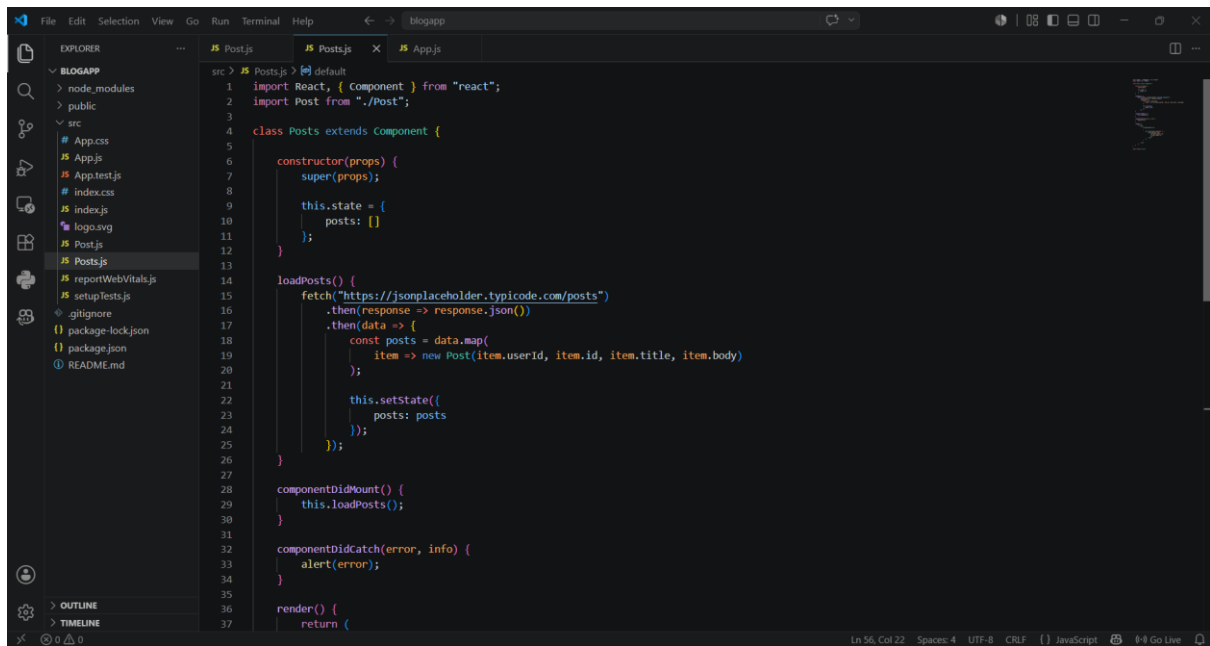


The screenshot shows the Visual Studio Code interface. The Explorer panel on the left displays the file structure of a project named 'BLOGAPP', including folders like 'node_modules', 'public', and 'src', and files like 'App.css', 'App.js', 'App.test.js', 'index.css', 'index.js', 'logo.svg', 'Post.js', 'Post.test.js', 'reportWebVitals.js', 'setupTests.js', '.gitignore', 'package-lock.json', 'package.json', and 'README.md'. The main editor area shows the 'App.js' file with the following code:

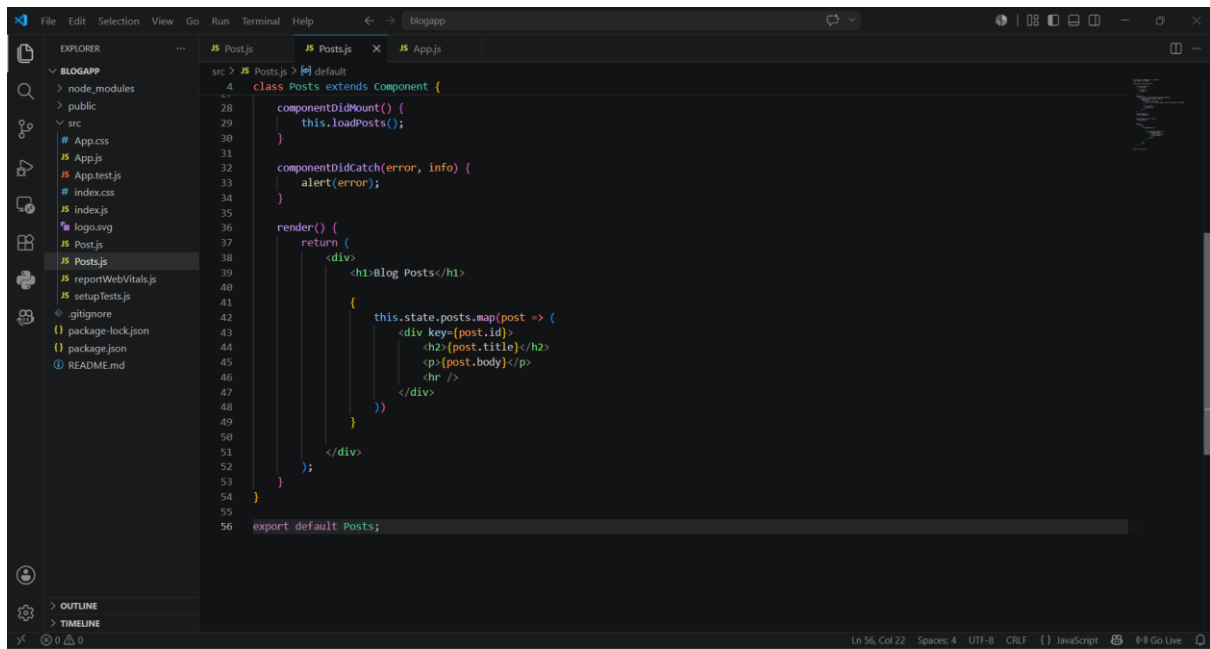
```
1 import React from "react";
2 import Posts from "../Posts";
3
4 function App() {
5   return (
6     <div>
7       <Posts />
8     </div>
9   );
10 }
11
12 export default App;
```

At the bottom, the Terminal panel shows the command 'PS C:\Users\jyots\myfirstreact\scorecalculatorapp\blogapp> npm start' and the output: 'Note that the development build is not optimized. To create a production build, use npm run build.' followed by 'webpack compiled successfully'.

Posts.js



```
1 import React, { Component } from "react";
2 import Post from "./Post";
3
4 class Posts extends Component {
5
6   constructor(props) {
7     super(props);
8
9     this.state = {
10       posts: []
11     };
12   }
13
14   loadPosts() {
15     fetch("https://jsonplaceholder.typicode.com/posts")
16       .then(response => response.json())
17       .then(data => {
18         const posts = data.map(
19           item => new Post(item.userId, item.id, item.title, item.body)
20         );
21         this.setState({
22           posts: posts
23         });
24       });
25   }
26
27   componentDidMount() {
28     this.loadPosts();
29   }
30
31   componentDidCatch(error, info) {
32     alert(error);
33   }
34
35   render() {
36     return (
```

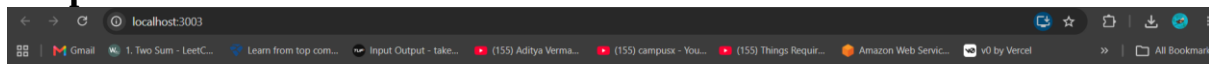


```
4 class Posts extends Component {
28   componentDidMount() {
29     this.loadPosts();
30   }
31
32   componentDidCatch(error, info) {
33     alert(error);
34   }
35
36   render() {
37     return (
38       <div>
39         <h1>Blog Posts</h1>
40         {
41           this.state.posts.map(post => (
42             <div key={post.id}>
43               <h2>{post.title}</h2>
44               <p>{post.body}</p>
45               <hr />
46             </div>
47           ))
48         }
49       </div>
50     );
51   }
52 }
53
54 export default Posts;
```

Post.js

```
1 class Post {
2   constructor(userId, id, title, body) {
3     this.userId = userId;
4     this.id = id;
5     this.title = title;
6     this.body = body;
7   }
8 }
9
10 export default Post;
```

Output



Blog Posts

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